

SHOP PRODUCT SALE AND BILLING SYSTEM

1. REQUIREMENTS

Requirement	Details
Product Details	Product ID, Product Name, Category and Price.
Stock Details	Stock ID, Product ID, Available Quantity and Stock Status.
Sale Details	Sale ID, Product ID, Sale Date, Quantity Sold, Sale Price and Total Amount.
Quantity	Enter required quantity, check availability and update stock after a successful sale.
Bill Calculation	$\text{Amount} = \text{Quantity} \times \text{Price}$; calculate subtotal, discount/tax if applicable and final amount.
Billing	Generate Bill ID, date, product details, quantity, price and final amount.
Functional	Manage products, stock, sales, quantity, bill calculation and billing.
Non-Functional	Accuracy, security, reliability, easy use, fast processing, backup and recovery.
Inputs	Product ID, quantity and sale/discount/tax information.
Outputs	Product details, stock, sale amount, updated stock and final bill.

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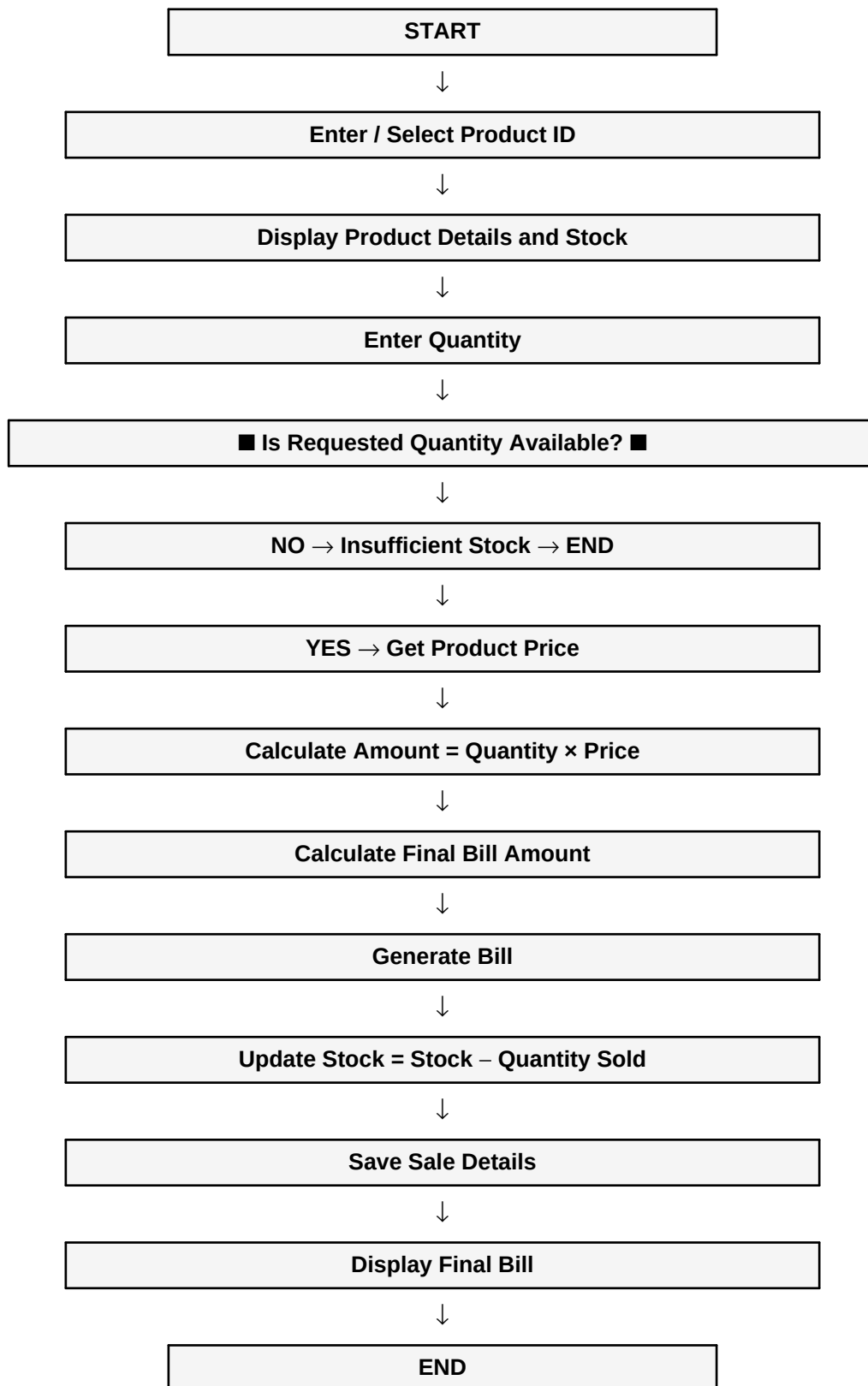
2. ALGORITHM

1. Start the system.
2. Enter or select Product ID.
3. Retrieve and display product details.
4. Display available stock quantity.
5. Enter required quantity.
6. Check whether requested quantity is available.
7. If quantity is unavailable, display "Insufficient Stock" and stop the transaction.
8. If available, retrieve product price.
9. Calculate Amount = Quantity × Price.
10. Calculate subtotal.
11. Apply discount and tax, if applicable.
12. Calculate final bill amount.
13. Generate the bill.
14. Update stock: New Stock = Available Stock – Quantity Sold.
15. Save sale details.
16. Display final bill.
17. Stop.

Formula: Amount = Quantity × Price

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3. FLOWCHART



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4. ER DIAGRAM

PRODUCT
PK Product_ID
Product_Name
Category
Price

STOCK
PK Stock_ID
FK Product_ID
Quantity_Available
Stock_Status

PRODUCT 1 : 1 STOCK

PRODUCT
PK Product_ID
Product_Name
Category
Price

SALE
PK Sale_ID
FK Product_ID
Sale_Date
Quantity_Sold
Sale_Price
Total_Amount

PRODUCT 1 : M SALE

SALE
PK Sale_ID
FK Product_ID
Sale_Date
Quantity_Sold
Sale_Price
Total_Amount

BILL
PK Bill_ID
FK Sale_ID
Bill_Date
Subtotal
Discount
Tax
Final_Amount

SALE 1 : 1 BILL

PK = Primary Key FK = Foreign Key